SHREYA THAKUR

+1-716-510-2084 | thakurshreya073@gmail.com | github.com/shreya7code | linkedin.com/in/shreya-thakur7/ | Portfolio | TX, USA (Open to All)

AI Software Engineer with 4+ years building Python and Java microservices, agentic AI, LLM, and RAG applications on AWS and Azure. Skilled in multi-agent orchestration, LLM evaluation, Kubernetes, Terraform, CI/CD, and MLOps.

SKILLS

Languages & Frameworks: Python, Java, TypeScript, SQL, FastAPI, REST APIs, Flask, Spring Boot, React, Next.js, Numpy, Pandas
Agentic AI & LLMs: LangChain, LangGraph, Multi-Agent Orchestration, RAG, Prompt Engineering, LLM Evaluation, LLM-as-Judge, Pinecone
Machine Learning & MLOps: PyTorch, TensorFlow, Scikit-learn, Transformer models, NLP, Fine-Tuning, MLflow, Model Serving & Deployment
Cloud & DevOps: AWS (EC2, ECR, ECS Fargate, Lambda, S3, Bedrock, SageMaker), Azure (Functions, AI Search, DevOps, Key Vault), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Jenkins
Data & Distributed Systems: PostgreSQL, MongoDB, Tableau, InfluxDB, Redis, Kafka, Celery, PySpark, Databricks, Azure Data Factory
Engineering Practices: Microservices, System Design, JWT/OAuth2, OWASP, pytest, JUnit, Prometheus, Grafana, Agile Scrum, Code Reviews
Certifications: AWS Solutions Architect – Associate, Introduction to Generative (AWS), AI-900 (Microsoft), DBMS (NPTEL), Python (MTA)

WORK EXPERIENCE

Sayaam for All | Full Stack Developer | CA, USA *Aug 2025 – Jun 2026*

- Built LLM microservices and model-serving endpoints with **Python**, **FastAPI**, async **REST APIs**, and **MongoDB**, boosting scalability 40%.
- Built zero-shot intent routing with Hugging Face **BART-MNLI** Transformers for **JWT/OAuth2** multi-tenant clients, raising accuracy 25%.
- Built **RAG** workflows with **Pinecone**, **LangChain**, and **LangGraph**, integrating **OpenAI**, **Amazon Bedrock**, and self-hosted **Ollama** inference for privacy, reducing external API costs by 30%.
- Deployed ML services via **Docker**, **Kubernetes**, **GitHub Actions** CI/CD, and **pytest** on **AWS EC2/ECR** with **Terraform**, cutting setup 70%.
- Built safety guardrails aligned to **OWASP Top 10 for LLM Applications**: prompt injection scanning, **Presidio** PII redaction, and **LLM-as-judge** red-teaming, detecting ~90% of attacks.
- Tracked p95/p99 latency & TTFT in **Prometheus** & **Grafana** with **LLM evals** (Recall@K, groundedness), fixing a production latency regression.

University at Buffalo | Graduate Teaching Assistant | NY, USA *Sep 2024 – Jun 2025*

- Mentored 200+ graduate students in deep learning, reviewing **PyTorch** and **Apache Spark** code on **CNNs**, **Transformers**, & model evaluation.
- Supported applied research on an end-to-end document intelligence system for OCR, summarization, translation, and TTS, integrating **mT5**, **NLLB**, and **FastText** language models.
- Architected a custom OCR pipeline with **ResNet**, **CRNN**, **BiLSTM**, and **CTC**, achieving 92% accuracy and 35% CER reduction with GPU inference, deployed via **Docker** on **Azure Container Registry** and **Azure App Service**.
- Fine-tuned **TrOCR** with a custom tokenizer for multilingual text, tracking **CRNN** checkpoints and CER in **MLflow** for model selection.

Wipro Ltd. | Software Engineer - Backend | Kolkata, India *Aug 2019 – Aug 2022*

Project: British Petroleum (BP)

- Programmed an AI-powered semantic search platform using **Python**, **Azure AI Search**, **NLP**, and a **React** frontend, indexing 10M+ documents, improving search precision by 25% and reducing query latency by 40%.
- Developed ML-based PII detection microservices using **Flask**, **Presidio**, event-driven **Azure Functions**, and **Kafka** consumers, reducing compliance effort by 80%.
- Built an NLP preprocessing microservice with **Python** and **NLTK** for **Azure AI Language** entity recognition, improving ML accuracy by 20% via tokenization and normalization.
- Engineered **Azure Data Factory** and **PySpark** pipelines on **Azure Databricks** and **Azure SQL**, delivering an audit analytics dashboard in 10 days, processing 10TB+ with 35% lower latency.
- Improved CI/CD with **Docker**, **Azure DevOps**, and **Key Vault** in **Agile** sprints on **Azure Boards**, cutting deployment failures 30% via PR reviews.

Project: Cisco Partner Portal

- Built scalable backend services with **Java**, **Spring Boot**, **SQL**, and **REST APIs** for partner onboarding, cutting API latency by 35%.
- Refactored backend modules with **OOP** design, **Spring Data JPA**, and **Hibernate**, improving maintainability and distributed delivery by 40%.
- Automated functional, API, & integration testing - **Selenium**, **Cucumber**, **Karate**, & **JUnit**, achieving 95% test coverage & reducing regressions.
- Integrated CI/CD pipelines with **Jenkins**, **Docker**, and **Maven**, reducing release cycles by 60% and enabling faster delivery of backend features.
- Mentored new engineers & authored onboarding docs adopted as team standard, conducting PR code reviews in **Agile** Scrum sprints via **Jira**.

PROJECTS

AgenticAI Model Safety Evaluation Platform [Link] *Jan 2026*

- Developed an agentic AI safety and evaluation platform automating LLM red-teaming for **OWASP Top 10 for LLM** risks across 1,500+ test cases, reducing manual review by 90% with **Grafana** dashboards.
- Built a 4-agent **LangGraph** pipeline with **TypeScript**, **Next.js**, **Celery**, **Redis**, **pgvector**, & **PostgreSQL** for A/B tests and **LLM-as-judge** scoring.

Smart Infra Monitor – Real-Time Infra Monitoring & Anomaly Detection [Link] *Jul 2025*

- Built ML anomaly detection with **Scikit-learn** IsolationForest, processing 2K+ events/sec via event-driven **Kafka**, **FastAPI**, & **InfluxDB** pipelines.
- Deployed **Docker** microservices on **AWS ECS Fargate** with a **React** dashboard, using rule-based and ML detection to boost monitoring 30%.

EDUCATION

University at Buffalo, The State University of New York | MS (Masters): Computer Science and Engineering *Buffalo, NY*
West Bengal University of Technology | B.Tech (Bachelors): Information Technology *Kolkata, WB, India*